

OpenStack Load Balancer as a Service (LBaaS): The Role of Resources

In this article, we compare the time taken and success rates while running different Rally tasks on load balancers using the providers Amphora and OVN for OpenStack, and outline the reasons for failures (if any). This can help administrators choose the right provider driver, depending on the service and resource requirements. Results have shown that OVN load balancers can be created and used in environments with low resources, while creating Amphora load balancers resulted in errors due to lack of resources.



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OpenStack is a cloud computing infrastructure that manages the compute, storage and networking resources of a data centre. It also provides a dashboard to help users provision resources with the help of a Web interface. The dashboard also gives administrators control over the management of resources. Besides Infrastructure-as-a-Service (IaaS), OpenStack provides other functionalities like fault management, orchestration, load balancing and service management.

A load balancer is used to allocate resources to various tasks to make the overall processing efficient. Load Balancer as a Service (LBaaS) is implemented as a networking service in OpenStack. So it uses a port in Neutron

(Networking-as-a-Service) and is assigned an IP from a subnet. Each port that the load balancer uses is specified by a listener, through which it can listen for requests. A listener is used to monitor the communications between the virtual machines and OpenStack services.

Load balancers use health monitors to divert traffic away from members (servers with specific IPs and ports) that do not respond properly. Each health monitor is associated with a pool — a list of members serving content through the load balancer.

If there are no load balancers, newly created virtual servers won't receive incoming traffic properly or not receive it at all. The resource allocation also becomes unbalanced when some virtual servers do not receive any traffic while some are overloaded.

The algorithms used for load balancing are ROUND_ROBIN, LEAST_CONNECTIONS, and SOURCE_IP.

- **ROUND_ROBIN:** Nodes are selected on a round-robin basis (each node is used for a fixed time period) for workload handling; so all nodes are under equal pressure to handle workloads.
- **LEAST_CONNECTIONS:** Nodes are selected in increasing order of the number of established connections from the client.
- **SOURCE_IP:** Hash values are computed based on the IP addresses of the clients and the server that is used for routing, to ensure that the requests from a client always go to the server with which the connection was